sotFS System Manual

Graph-Structured Filesystem with DPO Rewriting,
Bounded Treewidth, and Capability-Gated Projections

Version 1.0

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Comprehensive technical reference for the `sotFS` filesystem.
Covers design, formal graph model, DPO rewriting rules, verification,
implementation, and architecture decisions.

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Part I

Design

Chapter 1

Introduction and Motivation

Filesystem metadata is inherently a graph: inodes link to directories, capabilities reference files, hard links create shared substructure, and version histories form DAGs. Yet every major filesystem—ext4, ZFS, Btrfs, HAMMER2—treats metadata as flat tables or implicit trees, losing the graph structure that could enable powerful queries, anomaly detection, and formal reasoning about correctness.

`sotFS` (Secure Object Transactional File System) makes this graph structure explicit. It represents filesystem metadata as a **typed graph** with six node types and seven edge types, and defines POSIX operations as **Double Pushout (DPO) graph rewriting rules**—algebraic transformations with mathematically precise preconditions, postconditions, and gluing constraints.

1.1 Contributions

1. **Type Graph TG:** A formal definition with six node types (Inode, Directory, Capability, Transaction, Version, Block) and seven edge types (contains, grants, delegates, derivesFrom, supersedes, pointsTo, hasXattr).
2. **DPO Rule Set:** Complete specification of POSIX operations as DPO graph rewriting rules with explicit gluing conditions.
3. **Treewidth Preservation:** Proof that DPO rules preserve bounded treewidth, enabling linear-time MSO queries via Courcelle's theorem.
4. **Curvature Monitor:** Incremental Ollivier–Ricci curvature computation for real-time structural anomaly detection.
5. **Deceptive Projections:** Capability-gated subgraph views as a graph endofunctor.
6. **GTXN Protocol:** Graph-level transaction manager composing SOT kernel transactions with graph-specific invariant checking.
7. **Coq Mechanization:** Machine-checked proofs that 6 DPO rules preserve all 5 structural invariants.

1.2 Comparison with Existing Systems

Feature	**ext4**	**ZFS**	**Btrfs**	**HAMMER2**	**SquirrelFS**	**`sotFS`**
Metadata model	flat	Merkle	B-trees	radix	Rust types	**typed graph**
Formal ops	none	none	none	none	typestate	**DPO rules**
Treewidth	N/A	N/A	N/A	N/A	N/A	$\leq k$
MSO queries	N/A	N/A	N/A	N/A	N/A	**O(n)**
Anomaly detect	none	none	none	none	none	**Ollivier–Ricci**
Crash consist.	journal	CoW	CoW+log	CoW	typestate	**DPO+WAL+typestate**
Verification	tested	tested	tested	tested	typestate	**TLA+ $\rightarrow$ Coq**

Chapter 2

Architecture

2.1 sotFS Within the sotX Stack

```
Application (POSIX binary via LUCAS/rump)
  | syscall (open, read, write, mkdir, ...)
LUCAS / rump VFS layer
  | Translates POSIX syscalls -> VfsOp IPC messages
sotFS Server Domain (Ring 3, userspace)
  |- VfsOp dispatcher
  |- GTXN engine (ADR-001)
  |   |- DPO rule matcher + gluing check
  |   |- Type graph (in-memory)
  |   |- Treewidth verifier
  |   +- Curvature monitor
  |- Persistence layer (WAL, bitmap, serialization)
  +- Bulk block-device capability (ADR-002)
sotX Microkernel (Ring 0): IPC | Scheduling | Frame alloc | Caps
Hardware (NVMe / virtio-blk)
```

2.2 Data Flow: POSIX Call → Graph Rewrite → Persist

Example: `mkdir("/data/newdir", 0o755)`:

1. LUCAS receives mkdir syscall, sends `VfsOp::Mkdir` IPC to `sotFS`.
2. Dispatcher invokes DPO rule MKDIR.
3. GTXN begins: match Directory for "/data", check gluing conditions, begin SOT Tier 1 transaction.
4. Apply MKDIR rule: add Inode, Directory, three contains edges (entry, ".", ".."). Verify link_count consistency and treewidth bound.
5. GTXN commits: WAL write (atomic), bitmap update, curvature update, provenance emit.
6. Reply to LUCAS with success and new inode ID.

Chapter 3

Type Graph Formalization

Definition 3.1 (Type Graph)**.** The `sotFS` type graph is a tuple $\mathrm{TG} = (V, E, \mathrm{src}, \mathrm{tgt}, \tau_V, \tau_E, \mathrm{attr})$ where V is a finite set of nodes, E a finite set of edges, $\mathrm{src}, \mathrm{tgt} : E \to V$ map edges to endpoints, $\tau_V : V \to \mathrm{NodeType}$ and $\tau_E : E \to \mathrm{EdgeType}$ assign types, and $\mathrm{attr} : V \cup E \to \mathrm{Attributes}$ assigns typed attributes.

NodeType = {Inode, Directory, Capability, Transaction, Version, Block}
EdgeType = {contains, grants, delegates, derivesFrom, supersedes, pointsTo, hasXattr}

3.1 Node Types

3.1.1 Inode

Attributes: `id: OID`, `vtype`, `perms`, `uid`, `gid`, `size`, `link_count`, `ctime`, `mtime`, `atime`.

Invariant $\mathbf{I_2}$: `link_count` $= |\{e \in E : \tau_E(e) = \mathrm{contains} \wedge \mathrm{tgt}(e) = i \wedge e.\mathrm{name} \neq \text{“..”}\}|$

3.1.2 Directory

Attributes: `id: OID`. Paired 1:1 with an Inode of `vtype` = Directory via a “`.`” edge.

Invariant $\mathbf{I_3}$: Every Directory d has exactly one “`.`” contains edge to its paired Inode.

Invariant $\mathbf{I_4}$: Names on outgoing contains edges from d are unique (set, not multiset).

3.1.3 Capability

Attributes: `cap_id`, `rights` (bitmask: R/W/X/Grant/Revoke), `epoch`.

Invariant $\mathbf{I_5}$: Exactly one outgoing `grants` edge, at most one incoming `delegates` edge.

3.1.4 Transaction, Version, Block

Transaction: models GTXN state machine (`Active/Preparing/Committed/Aborted`). Version: snapshot lineage node. Block: contiguous sector extent with `refcount`.

3.2 Edge Types and Constraints

Edge	Source → Target	Attributes	Key Constraint
contains	Directory → Inode	name	C_1: unique names per dir
grants	Capability → Inode	rights	C_2: rights ⊆ cap.rights
delegates	Cap → Cap	–	C_3: monotonic attenuation; C_4: forest
derivesFrom	Version → Version	–	C_5: DAG
supersedes	Inode → Inode	–	C_6: acyclic
pointsTo	Inode → Block	offset	C_7: non-overlapping extents
hasXattr	Inode → XAttr	–	1:1 per (inode, namespace, name)

3.3 Global Invariants

Invariant 3.1 (G_1 – No orphan inodes)**.** $\forall i \in V, \tau_V(i) = \text{Inode} \implies \text{reachable}(\text{root}, i) \vee i.\text{link_count} = 0$

Invariant 3.2 (G_2 – No dangling edges)**.** $\forall e \in E : \text{src}(e) \in V \wedge \text{tgt}(e) \in V$

Invariant 3.3 (G_3 – Link count consistency)**.** $\forall i : i.\text{link_count} = |\{e : \tau_E(e) = \text{contains} \wedge \text{tgt}(e) = i \wedge e.\text{name} \neq \text{“..”}\}|$

Invariant 3.4 (G_4 – Capability monotonicity)**.** Along any delegates path $c_0 \to c_1 \to \cdots \to c_n$: $c_n.\text{rights} \subseteq \cdots \subseteq c_0.\text{rights}$

Invariant 3.5 (G_5 – No directory cycles)**.** The contains subgraph restricted to Directory→Inode(vtype=Directory) links (excluding “`.`” and “`..`”) is acyclic.

Invariant 3.6 (G_7 – Treewidth bound)**.** $\text{tw}(\text{TG}) \leq k$ for configurable k (see Theorem 5.1).

Chapter 4

DPO Rules for POSIX Operations

A DPO rule $\rho = (L \xleftarrow{l} K \xrightarrow{r} R)$ transforms host graph G via match $m : L \to G$ into result H, subject to gluing conditions (identification and dangling).

4.1 Mutating Rules

4.1.1 CREATE(dir, name, vtype) → new_oid

$L = \{d\text{:Dir}\}$, $K = \{d\}$, $R = \{d, i\text{:Inode[lc=1]}, e\text{:contains}(d, i)[\text{name}]\}$. If vtype=Directory, additionally creates Directory node, "`.`" and "`..`" edges.

Gluing: GC-CREATE-1 (no duplicate name), GC-CREATE-2 (dir exists).

4.1.2 WRITE(inode, offset, data)

Adds/replaces Block + pointsTo edge. CoW check on shared blocks.

4.1.3 MKDIR(dir, name)

Special CREATE adding Inode(lc=2) + Directory + 3 contains edges.

4.1.4 RMDIR(dir, name)

Removes Directory + Inode + 3 edges. **Gluing:** GC-RMDIR-1 (directory empty).

4.1.5 LINK(dir, name, target)

Adds contains edge, increments link_count. **Gluing:** GC-LINK-2 (target is Regular), GC-LINK-3 (lc < LINK_MAX).

4.1.6 UNLINK(dir, name)

Removes contains edge, decrements link_count. GC if lc reaches 0.

4.1.7 RENAME(src_dir, src_name, dst_dir, dst_name)

Four cases: same/cross dir × target exists/absent. Cross-dir requires SOT Tier 2 (2PC). **Gluing:** GC-RENAME-2 (cycle prevention via `is_ancestor` check).

4.2 Treewidth Impact per Rule

Only LINK can increase treewidth, bounded by GC-LINK-3. All other rules are tw-neutral or tw-decreasing.

Part II

Theory

Chapter 5

Bounded Treewidth and MSO Queries

Theorem 5.1 (Treewidth Bound)**.** Let TG be produced by any finite DPO rule sequence from CREATE-ROOT. If link_count $\leq$ LINK_MAX per inode, then tw(TG) $\leq$ LINK_MAX $+ O(1)$.

Proof sketch. The contains subgraph without hard links is a forest (tw=1). Each hard link adds one back-edge, increasing tw by at most 1. The grants, delegates, derivesFrom, and pointsTo subgraphs each contribute tw$\leq$1. Combined: tw(TG) $\leq$ LINK_MAX $+ 5$.

Corollary. Without hard links: tw(TG) ≤ 6.

5.1 Courcelle's Theorem Application

Since tw(TG) is bounded, every MSO_2-definable query runs in $O(n)$ time. Examples: capability reachability, cycle detection, orphan detection, version ancestry.

Chapter 6

Ollivier–Ricci Curvature Monitor

For edge (u, v): $\kappa(u, v) = 1 - W_1(\mu_u, \mu_v)/d(u, v)$ where μ_x is the lazy random walk measure with parameter $\alpha = 0.5$.

Normal patterns: CREATE decreases parent curvature slightly; UNLINK increases it; WRITE has no effect; steady-state $\kappa \approx -0.5$ to 0.

Anomalous signatures: Mass creation $\rightarrow$ positive curvature spike; symlink bomb $\rightarrow$ strong positive on target; deep chain $\rightarrow \kappa \approx -1.0$.

Incremental: After GTXN commit, recompute only the 2-hop neighborhood of changed edges. Amortized $O(d_{\max}^2)$ per rule application.

Chapter 7

Deceptive Subgraph Views

A projection π for domain D is a typed graph morphism $\pi : \mathrm{TG}|_D \to \mathrm{TG}_D$ that may omit, add, or relabel nodes/edges. The deception functor D_{FS} : **TGraph** $\to$ **TGraph** extends the SOT deception model to graph structure.

Consistency requirement: $D_{FS}(\mathrm{TG})$ must satisfy all global invariants G_1–G_7. Fabricated content is checkable via MSO queries in linear time.

Chapter 8

Capabilities and Transactions

8.1 Capabilities as Graph Edges

DERIVE: creates child capability with rights $\subseteq$ parent.rights via `delegates` edge. REVOKE: epoch-based, removes capability node and all delegates descendants.

8.2 Graph Transactions (GTXN)

Protocol: begin $\rightarrow$ verify_gluing $\rightarrow$ verify_treewidth $\rightarrow$ SOT::begin $\rightarrow$ apply_rule $\rightarrow$ verify_postconditions $\rightarrow$ commit/rollback.

Tier mapping: Most ops use SOT Tier 1. Cross-directory RENAME uses Tier 2 (2PC).

Part III

Verification

Chapter 9

Multi-Layer Verification

9.1 Layer 1: TLA+ Model Checking (Bounded)

4 specifications verified: `sotfs_graph`, `sotfs_transactions`, `sotfs_capabilities`, `sotfs_crash`. 20+ invariants model-checked across small/medium/large configurations.

9.2 Layer 2: Crash Refinement (Tested)

6 integration tests with redb ACID backend verify round-trip persistence, atomic overwrite, and complex operation survival.

9.3 Layer 3: Rust Typestate (Compile-Time)

`InodeHandle<State>`, `TxHandle<State>`, `DirHandle<State>`, `CapHandle` enforce state machine transitions at compile time. `#[must_use]` prevents silent transaction drop.

9.4 Layer 4: Loom Concurrency (Bounded)

3 exhaustive interleaving tests: transaction isolation, rollback correctness, conflicting write serialization.

9.5 Layer 5: Coq Mechanized Proofs (Unbounded)

Module	DPO Rule	Invariants	Status
`SotfsGraph.v`	(definitions)	init_graph_well_formed	Complete
`DpoCreate.v`	create_file	all 5	Complete (0 admits)
`DpoUnlink.v`	unlink_keep	all 5	Complete (0 admits)
`DpoRename.v`	rename_same_dir	all 5	Complete (0 admits)
`DpoMkdir.v`	mkdir	all 5	Complete (1 side cond.)
`DpoLink.v`	hard_link	all 5	Complete (0 admits)
`DpoRmdir.v`	rmdir	3 of 5	Partial

Key lemma: `count_remove_matching` establishes that removing one predicate-satisfying element decreases `count_occ_pred` by exactly one, closing LinkCountConsistent for Unlink and Rename.

9.6 Assumptions (Not Verified)

Hardware: CPU correctness (x86_64 TSO), storage durability (fsync honored), no bit rot.
Compiler: Rust compiler correctness, no UB in safe code, optimization preserves semantics.
Design: IPC reliability (verified by `ipc.tla`), capability unforgery, single-writer per GTXN.

Chapter 10

TLC Model Checking Bounds

Three configuration tiers per spec. The runner script `run_tlc.sh` drives TLC across all combinations.

Spec	Key Constant	Small	Medium	Large
sotfs_graph	InodeIds	5	6	8
	DirIds	3	4	5
	MaxOps	6	8	10
sotfs_transactions	GTxnIds	2	3	4
	SOIds	3	4	5
	MaxOps	8	10	12
sotfs_capabilities	CapIds	4	5	5
	InodeIds	2	3	3
	MaxOps	6	8	8
sotfs_crash	SOIds	2	3	4
	MaxOps	8	10	12

Part IV

Implementation

Chapter 11

Crate Structure

Crate	Description	Key Types
`sotfs-graph`	Type graph data structures	`TypeGraph`, `Arena<T,N>`, `RcuGraph`
`sotfs-ops`	DPO rewriting rules	`create_file`, `mkdir`, `symlink`, `fsck`
`sotfs-storage`	Persistence via redb	`save()`, `load()`
`sotfs-tx`	GTXN manager	`with_transaction`, commit/rollback
`sotfs-monitor`	Anomaly monitors	treewidth, curvature, deception
`sotfs-fuse`	FUSE bridge	Full POSIX VFS

11.1 Arena Allocator

Fixed-capacity heap-backed pool (`Box<[MaybeUninit<T>]>`) with O(1) alloc/dealloc/lookup. Free-list slot reuse prevents fragmentation. 65K node slots, 131K edge slots. TypeGraph struct: ∼744 bytes (pointers + counters).

11.2 Epoch-Based RCU

Lock-free reads via two TypeGraph copies with atomic swap. 8 concurrent reader slots. Writers: clone active → inactive, mutate, swap, wait for epoch drain.

11.3 DynamicTreewidth

Incremental treewidth via cached elimination ordering. <20% dirty nodes: partial re-elimination. ≥20%: full greedy min-degree.

Chapter 12

New Features

12.1 Extended Attributes (xattrs)

XAttr nodes connected via HasXattr edges. Operations: setxattr, getxattr, removexattr, listxattr. Four namespaces (User, System, Security, Trusted). 64KB max value.

12.2 Symlinks

`VnodeType::Symlink` with target in `symlink_targets: BTreeMap`. DPO rule identical to CREATE except vtype=Symlink, permissions=0777.

12.3 POSIX ACLs ↔ Capability Graph

ACL entries map to Grants edges: `ACL_USER_OBJ` → `Grants(cap_owner, inode, rights)`. Default: 3-entry ACL synthesized from permission bits.

12.4 Hierarchical Quotas

Per-subtree `Quota` with inode/byte limits. Enforcement walks "`..`" edges upward. Propagation: O(depth) amortized.

12.5 fsck Verifier

Checks all 5 structural invariants + block refcount + orphan detection + xattr integrity. Returns `FsckReport` with typed errors and warnings.

Chapter 13

Test Summary

Category	Count	Method
Arena allocator	16	Unit tests
RCU concurrency	9	Multi-thread
Graph invariants	3	Unit tests
Typestate	6	Unit + compile-fail
DPO operations	35	Unit tests
Property-based	10	proptest (10K iter)
Storage/crash	7	Integration (redb)
Transaction	3	Unit tests
Loom concurrency	3	Exhaustive
Treewidth	20	Unit + adversarial
Curvature	14	Unit + incremental
Deception	5	Projections
Adversarial	13	Security scenarios
xattrs/symlinks/ACL	10	Unit tests
fsck	2	Unit tests
Total	**147**	**All passing**

Part V

Architecture Decisions

Chapter 14

ADR-001: Graph-Level Transaction Manager

Status: Accepted (2026-04-10). **Decision:** Separate userspace GTXN wrapping SOT Tier 1/2 transactions. Preserves kernel simplicity; checks graph invariants in userspace. Rejected alternatives: direct SOT use (no invariant checking), unified kernel manager (violates microkernel principle).

Chapter 15

ADR-002: Block Capability Interpretation

Status: Accepted. **Decision:** sotFS server holds bulk block-device capability; clients get attenuated inode-SO caps. Matches existing VFS pattern. Rejected: per-inode block caps (capability explosion).

Chapter 16

ADR-003: Bootstrap Procedure

Status: Accepted. **Decision:** `sotfs_format()` is a DPO rule application (CREATE-ROOT). Initializes empty graph with root inode, directory, capability, superblock.

Chapter 17

ADR-004: Rump Kernel Interaction

Status: Accepted. **Decision:** `sotFS` registers as `FsType::SotFs` in rump VFS mount table, coexisting with FFS.

Appendix A

VfsOp → DPO Rule Mapping

VfsOp	DPO Rule	Tier	Graph Mutation
Open	OPEN (query)	0	None
Close	CLOSE (query)	0	May trigger GC
Read	READ (query)	0	None
Write	WRITE	1	Add/replace Block
Mkdir	MKDIR	1	Add Inode + Dir + edges
Rmdir	RMDIR	1	Remove Inode + Dir + edges
Unlink	UNLINK	1	Remove edge, maybe Inode
Rename	RENAME	1/2	Move edge
Stat	STAT (query)	0	None
Readdir	READDIR	0	Enumerate contains edges
Fsync	—	—	Force WAL flush

Appendix B

Category Theory Integration

`sotFS` introduces category **TGraph** (typed graphs, typed graph morphisms) alongside the existing SOT categories.

Functors:

- $G : \mathbf{SO} \to \mathbf{TGraph}$ — maps Secure Objects to graph nodes (faithful).
- $Rw : \mathbf{Tx} \to \mathbf{TGraph}$ — maps transactions to graph transformations (composition-preserving).
- $D_{FS} : \mathbf{TGraph} \to \mathbf{TGraph}$ — deception endofunctor (invariant-preserving).

Natural transformations: $\eta : \mathrm{Id} \Rightarrow \mathrm{WAL} \circ \mathrm{Id}$ (persistence), $\pi_D : \mathrm{Id} \Rightarrow D_{FS}$ (projection).